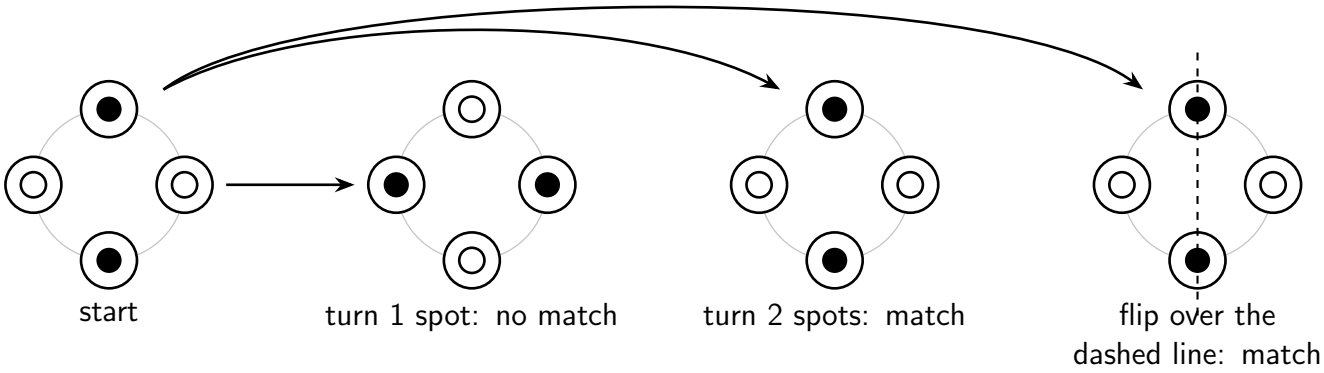
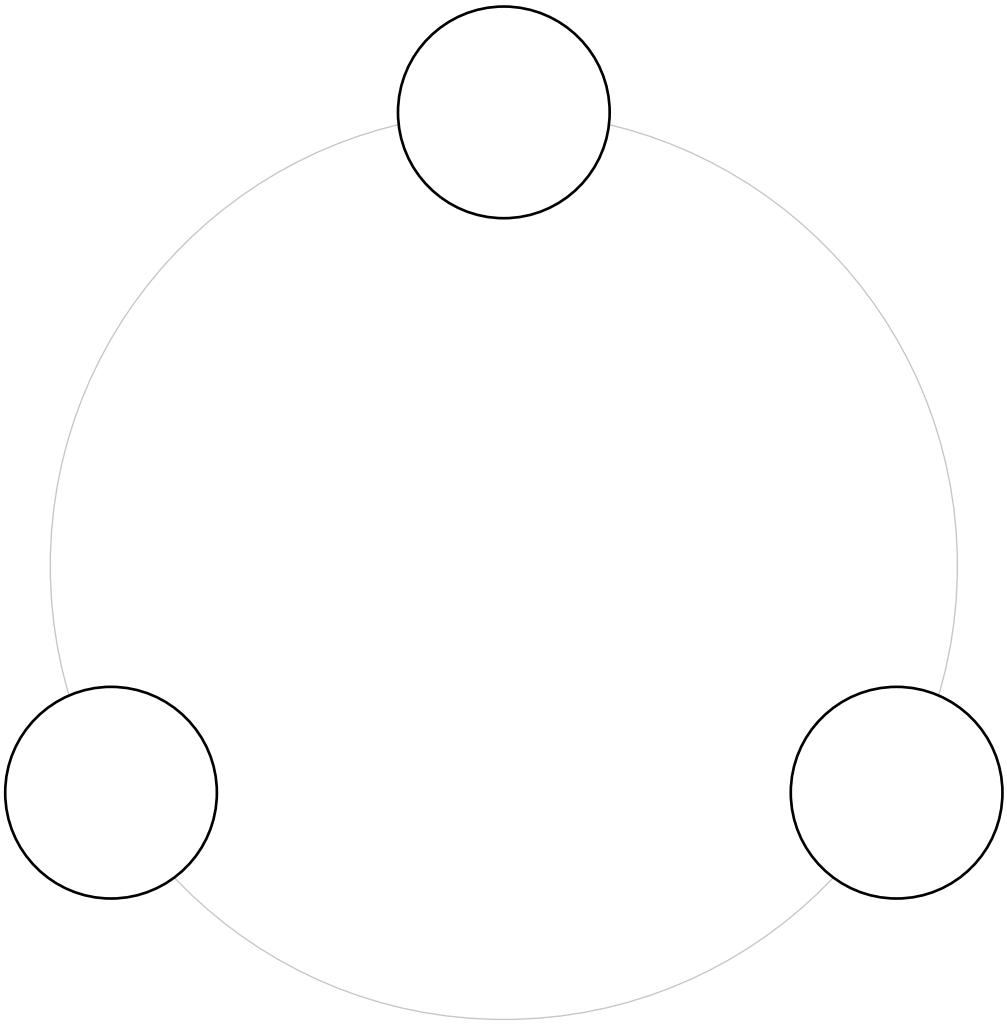
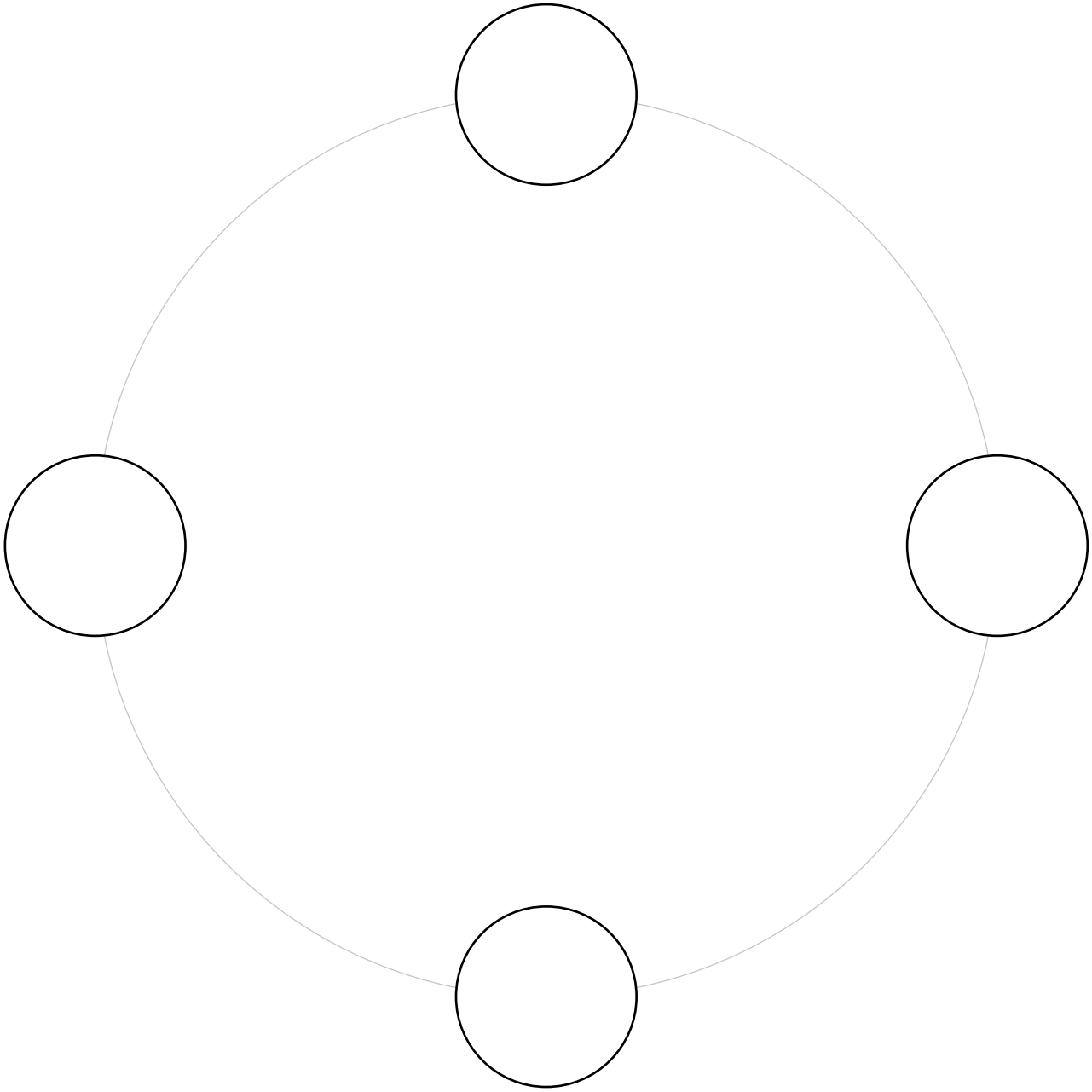
Fill every spot; neighbors may be the same kind. Only counter kinds and spots count. Keep the original still. Turn or flip (turn over) its tracing copy until all spots line up. A whole turn does not count.



Problem 1: Make different patterns on this ring with two kinds of counter. Let your partner find turns or flips that match. Can you make one your partner cannot match?

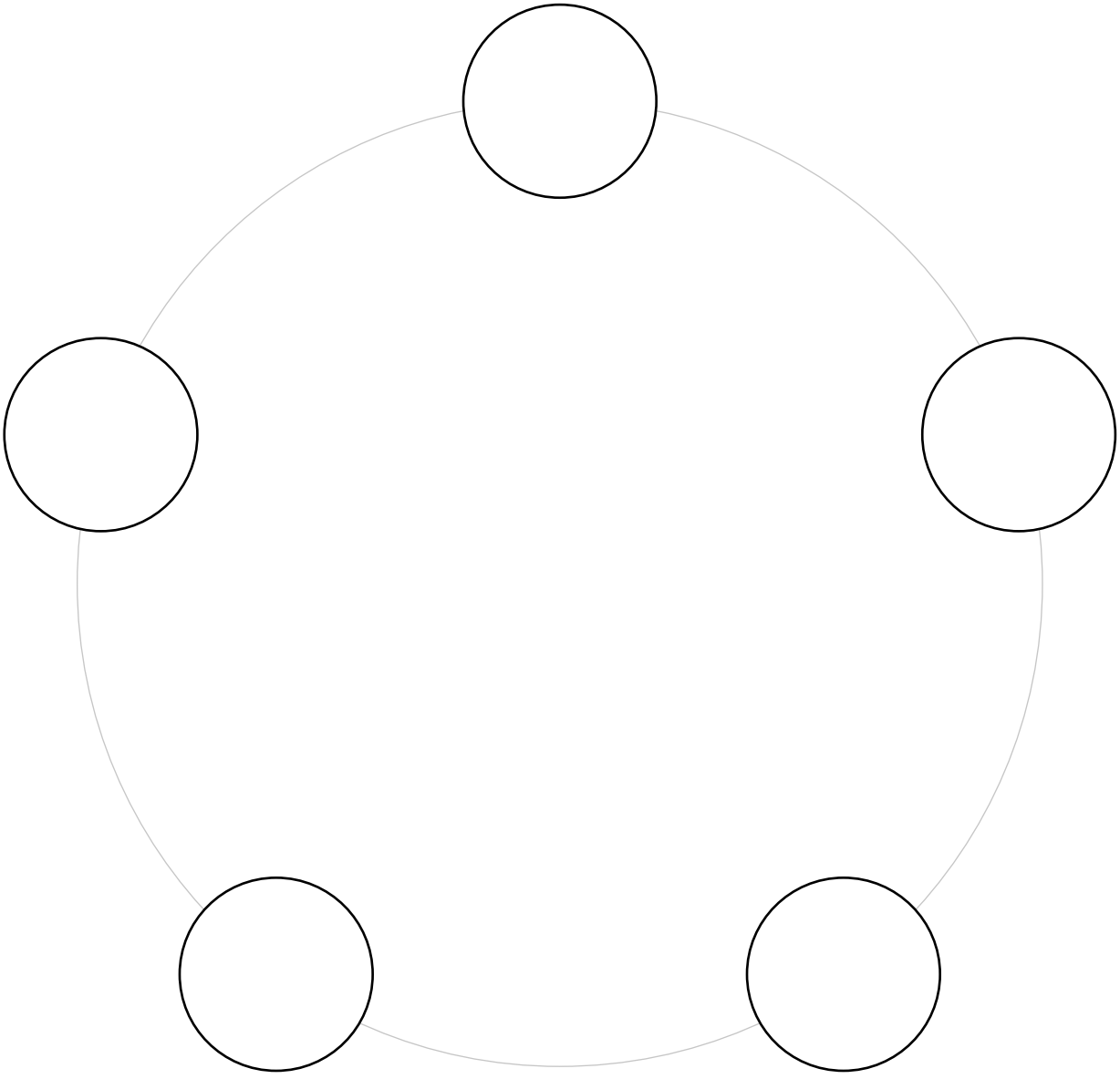


Problem 2: Make a pattern on this ring that no turn or flip can match. How few kinds of counter can you use?

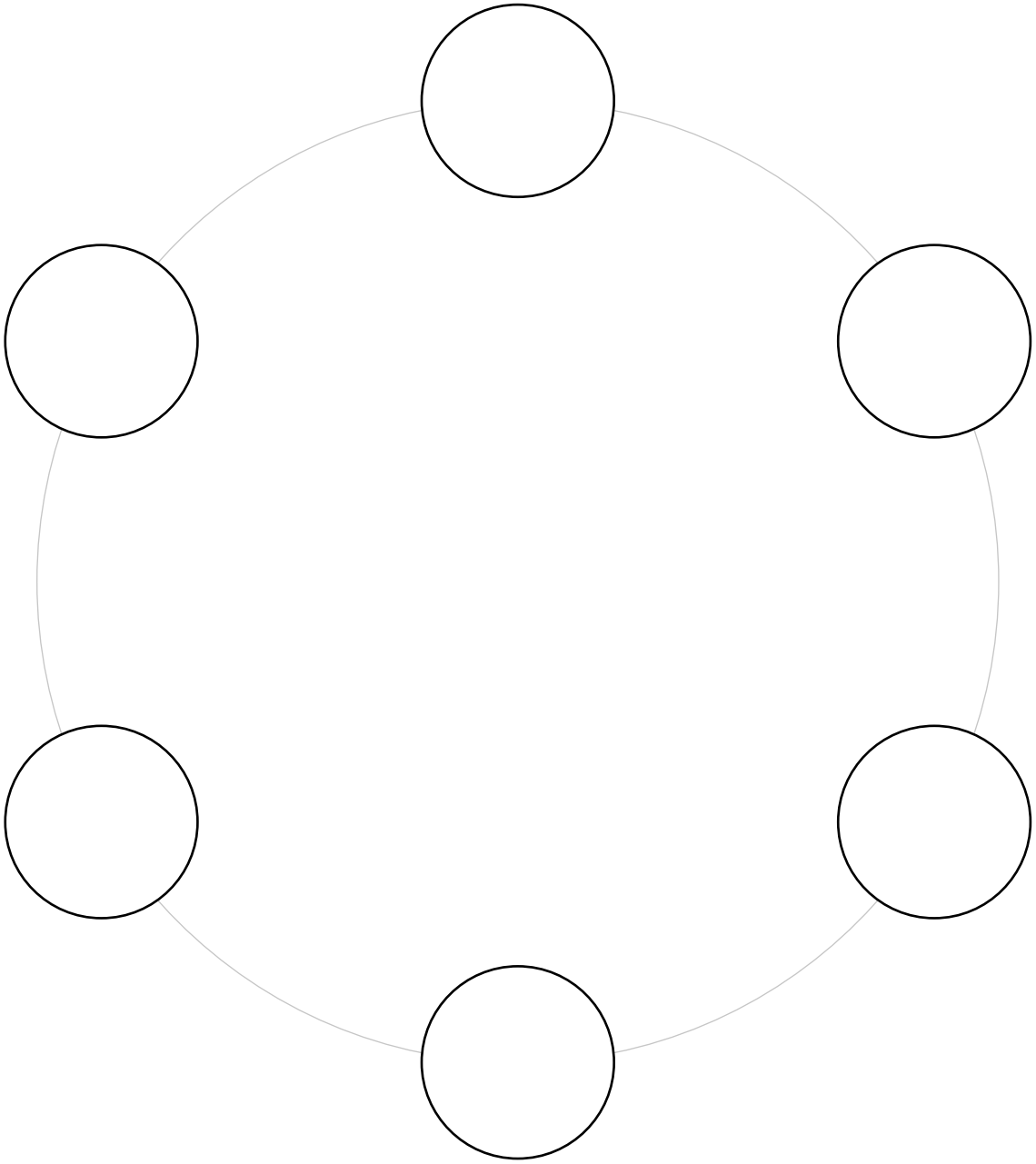
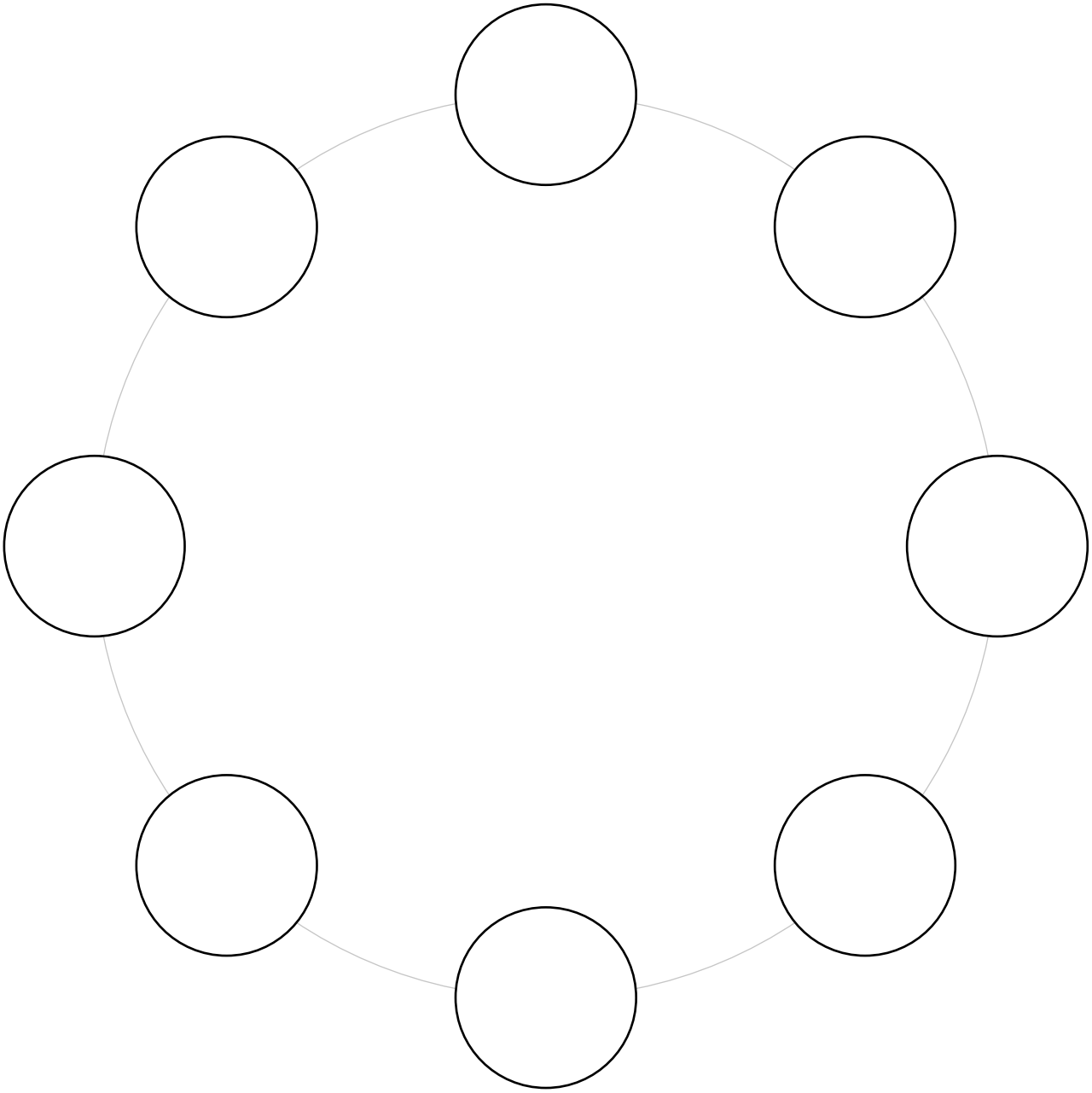


Problem 3: Change just one counter in different ways. Find changes that let some turn or flip match and changes that do not.

Problem 4: Can two kinds of counter make a pattern on this ring that no turn or flip can match? Try three kinds too.



Problem 5: Make a pattern with two kinds of counter that no turn or flip can match. Let your partner test it.



Problem 6: Build two two-kind patterns that no turn or flip can match and that cannot match each other. Keep each on its own tracing copy.